



3.8.7 Wrap-Up: Ticket Out the Door

1. The table shows the elevation, in meters (m), of five locations.

Location	Elevation (m)
Vpadina Kaundy, Kazakhstan	-132
Laguna del Carbón, Argentina	-105
Zuidplaspolder, Netherlands	-7
Fort Myers, Florida	2
Britton Hill, Lakewood, Florida	105

For each pair, use $<$, $>$, or $=$ to compare their distances from sea level.

Zuidplaspolder, Netherlands	____ ____ ____
Fort Myers, Florida	

Britton Hill, Lakewood, Florida	____ ____ ____
Laguna del Carbón, Argentina	

Laguna del Carbón, Argentina	____ ____ ____
Vpadina Kaundy, Kazakhstan	

Unit 3, Lesson 9: Operations With Absolute Value in the Real World



Benchmark

MA.6.NSO.1.4 Solve mathematical and real-world problems involving absolute value, including the comparison of absolute values.



Learning Target

- ☐ I can perform operations that involve the absolute value of two numbers.



3.9.1 Warm-Up: Math Fail

1. What do you notice about the sign?



3.9.2 Collaboration: Absolute Value and Profit

1. Landon starts a lawn-mowing business to earn money. All of his profits are being saved for college.

Profit is the difference between the amount earned (revenue) and the amount spent.

His father lends him a lawnmower to use for free, but he must pay for the gas and other supplies. For each lawn, his expenses are \$7.30, and he earns an average of \$40.00.

- a. In this situation, revenue is represented with a

- ☐ positive
☐ negative

value and an expense is represented

by a

- ☐ positive
☐ negative

value.

- b. Write a value to represent each situation.

Landon's Finances	Value
Expenses per lawn	
Revenue per lawn	

c. Use an inequality symbol, $<$ or $>$, to compare the absolute values of Landon’s expenses to revenue for each lawn he mows.

$$| \quad | > | \quad |$$

d. How does the inequality show that Landon will make a profit?

e. To make a profit, total expenses of a business must be

☐ greater than
 ☐ less than

 total revenue. A business is not profitable when the total

☐ expenses are
 ☐ revenue is

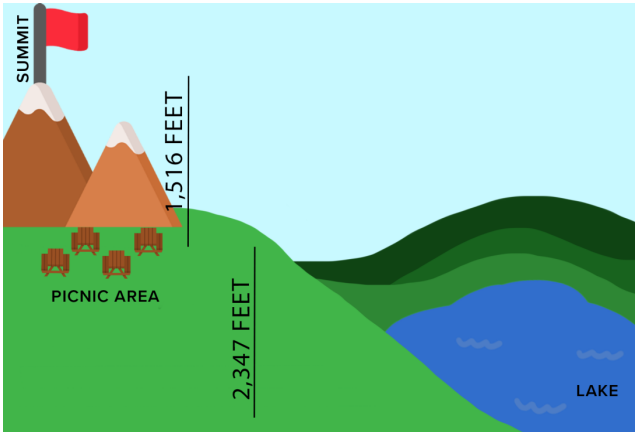
 greater than the

☐ expenses.
 ☐ revenue.



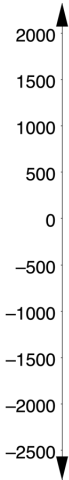
3.9.3 Guided Instruction: Operations With Absolute Value in the Real World

1. Javier and his family drove to a picnic area on a mountain. From the picnic area, the family hiked to the mountain’s summit, which was an elevation change of 1,516 feet (ft). They hiked back to the picnic area to eat lunch. After lunch, they hiked down to the lake, which was an elevation change of 2,347 ft.



a. In the table, write the value that represents the position of each location if the value of the position of the picnic area is 0. Then, plot each value at the estimated location on the number line.

Location	Value
Picnic area	0
Mountain summit	
Lake	



- b. Represent the distance of each location from the picnic area.

Location	Distance from Picnic Area
Mountain summit	
Lake	

- c. What operation would you use with the representations in the table to find the distance from the lake to the mountain summit?
- d. What is the distance from the lake to the mountain summit?
- e. What operation would you use to determine how much farther a hike to the lake is than a hike to the mountain summit?
- f. How much farther is a hike to the lake than a hike to the mountain summit?

2. Chloe is starting a painting business. The supplies she purchased for the business cost \$825. She will earn a fee of \$80 for each room that she paints. Chloe wants to determine how many rooms she will need to paint before she starts making a profit.

- a. ☐ Fees Chloe earns increase her profit.
☐ Supplies Chloe buys
- b. ☐ Fees Chloe earns decrease her profit.
☐ Supplies Chloe buys
- c. Write a value in the blank for each absolute value and a word in the remaining blank to complete each sentence.

To find the revenue, Chloe multiplies the number of rooms she paints by |____|.

To find the profit, Chloe subtracts |____| from the _____.

- d. In the table, complete the mathematical statement and then perform the operations to determine if the number of rooms painted would result in a profit.

Number of Rooms	Mathematical Statement	Profit (Yes or No)
6	6 ____ — ____	
10	10 ____ — ____	
13	13 ____ — ____	
16	16 ____ — ____	



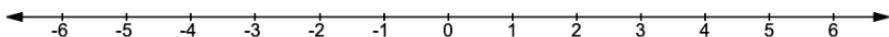


3.9.4 Try It: Operations With Absolute Value in the Real World

Logan, Mia, and Xavier all live on the same street as their school, which runs from east to west.

- Logan lives 4 blocks to the west of the school.
- Mia lives 5 blocks to the east of the school.
- Xavier lives 2 blocks to the west of the school.

1. Use the number line to model the position of each house and the school along the street. Label the location of each person's house and the school.



2. Work with your partner to write the mathematical sentence using absolute values. Then, use the mathematical sentence to determine each answer.
 - a. How far does Logan live from Mia?
 - b. How far does Mia live from Xavier?
 - c. How far does Xavier live from Logan?
 - d. Was the same operation used to find each distance?



3.9.5 Collaboration: Halfway Point

Mount Ojos del Salado is the highest mountain in Chile. Its peak is 6,893 meters above sea level. The lowest point of the Atacama Trench, just off the coast of Peru and Chile, is 8,065 meters below sea level.

Work with your partner to answer each question.

1. Determine how to label the number line so that you can approximate a location on the number line that represents Mount Ojos del Salado, Atacama Trench, and sea level.
2. Label the number line and place a point at the approximate location for Mount Ojos del Salado, Atacama Trench, and sea level.
3. Write a mathematical sentence using absolute value to find the difference in elevations between Mount Ojos del Salado and the Atacama Trench.



4. With your partner, discuss how the halfway point between the peak of Mount Ojos del Salado and the Atacama Trench can be found. Write down all of the strategies that were discussed.

5. Explain how to determine if the halfway point's elevation is above sea level or below sea level.



3.9.6 Your Turn!

1. The captain of a fishing vessel is standing on the deck at 23 feet above sea level. He holds a rope tied to a fishing net below him that is underwater at a depth of 38 feet.
- a. Draw a model of this situation using the number line.
- b. Write a mathematical sentence using absolute value to determine the length of the rope.



2. Aryan owes \$325.46 for recent purchases he made using his credit card, and he owes his parents another \$187.27.
- a. Write a mathematical statement using absolute value that can be used to determine Aryan's total debt.
- b. What is the value of Aryan's total debt?
3. When Brenda places a package of chicken in the freezer, the chicken is at a temperature of 41° Fahrenheit ($^{\circ}\text{F}$). While in the freezer, the temperature of the chicken changes -2°F every hour until it reaches the temperature of the freezer. What is the temperature of the chicken 3 hours after Brenda puts it in the freezer?



3.9.7 Wrap-Up: Reflection

1. Check all of the statements below that describe your understanding of and ability to work with absolute values.
- ☐ I can compare the absolute value of two numbers.
 - ☐ I need support on how to compare the absolute value of two numbers.
 - ☐ I can perform operations that involve the absolute value of two numbers.
 - ☐ I need support to perform operations that involve absolute value.
 - ☐ I can interpret comparisons of absolute values in real-world contexts.
 - ☐ I need support interpreting comparisons of absolute values in real-world contexts.
 - ☐ I can perform operations that involve absolute value when solving real-world problems.
 - ☐ I need support to perform operations that involve absolute value when solving real-world problems.